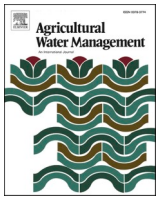
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Rainwater harvesting potential zone identification using GIS –Machine learning approach and comparison of their performance

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ABSTRACT

Rainwater harvesting (RWH) systems play a vital role in sustainable water management, yet their adoption is often constrained by the difficulty of accurately assessing site suitability and economic feasibility. This study aims to develop a machine learning (ML)–based geospatial framework to map RWH suitability in the Choke Watershed, Ethiopia, using twelve biophysical and hydrological conditioning factors. All spatial datasets were standardized and processed at a 30-m resolution, and the final suitability maps were generated at the same spatial scale to ensure consistent spatial comparability. Three ML algorithms—Random Forest (RF), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost)—were trained and validated using 175 suitable and 75 non-suitable inventory points, with a 70/30 training–testing split. Prior to model development, continuous predictors were normalized using min–max scaling, and all datasets were checked for missing values; incomplete DEM-derived pixels were corrected using bilinear interpolation, and unresolved records in thematic layers were removed to avoid bias during training. Model performance was evaluated using confusion-matrix metrics and the area under the receiver operating characteristic curve (AUC-ROC). The GB and XGBoost models showed the highest predictive accuracy (AUC = 0.97), followed by RF (AUC = 0.96). Spatial analysis revealed that highly suitable RWH zones are concentrated in flat terrain with loam and clay-loam soils and higher rainfall availability. This study represents the first application of GB and XGBoost models for RWH suitability mapping in the Choke Watershed and demonstrates the effectiveness of ML-driven geospatial modeling for identifying optimal RWH locations. The findings provide a valuable decision-support tool for planners and policymakers seeking to improve water resource management and climate-resilient agricultural development in the region.

1. Introduction

Water scarcity is one of the most pressing global challenges of the twenty-first century, particularly in developing nations where factors like population growth, climate change, water pollution, and unsustainable use of water resources are all contributing factors. Currently, about 3.6 billion people—over half of the global population—live in areas experiencing water shortages, and this number is projected to rise to 4.8–5.7 billion by 2050 (du Plessis, 2023). Despite water's abundance on Earth, only 2.5 % of total water is freshwater, with less than 1 % being accessible for human use (Mishra, 2023). Approximately one-third of global freshwater comes from groundwater, serving residential (36 %), agricultural (42 %), and industrial (27 %) needs (Musie and Gonfa, 2023; Tiwari et al., 2022).

Rainwater plays a critical role in replenishing groundwater resources. But climate change, urbanization, and too much water extraction have made it harder to find clean water around the world. To address this challenge, various strategies have been explored, with rainwater harvesting (RWH) emerging as an affordable, environmentally friendly, and adaptable solution (Chen et al., 2023; Rana and Moniruzzaman, 2023; Tolossa et al., 2020). RWH not only reduces surface runoff but also enhances groundwater recharge, supporting sustainable water management (Gebreslassie et al., 2025). Its effectiveness, however, depends on local hydrological, climatic, and socio-economic conditions (Al-Qatawneh et al., 2025).

In Africa, between 75 and 250 million people currently experience increased water stress, with rainfed agricultural yields projected to decline by up to 50 % in some regions (Gan et al., 2016; Tadele, 2017; Zhang et al., 2024). In Ethiopia, although the country is endowed with over 122 billion m^3 of annual surface runoff (Debebe et al., 2023a),

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water shortages remain acute due to uneven distribution, topographic constraints, and climate variability (Gebreeyesus et al., 2025; Tesfamariam et al., 2019). Smallholder farmers in particular face severe limitations in crop production because of irregular rainfall and limited irrigation infrastructure. As a result, RWH has emerged as a promising strategy for improving water accessibility and agricultural productivity (Chen et al., 2025; Kansal et al., 2025).

The hydrological and environmental circumstances in the Blue Nile Basin, particularly in the Choke basins, are ideal for RWH to enhance the water supply for residences and farmers (Gebremedhn et al., 2023). However, scientific site selection remains a significant challenge, requiring the integration of multiple spatial, hydrological, and socio-economic variables (Belachew et al., 2025).

Geospatial technologies, including Remote Sensing (RS), Geographic Information Systems (GIS), and Multi-Criteria Decision-Making (MCDM) techniques, have been extensively used worldwide to determine the best RWH locations (Ejegu and Yegizaw, 2020; Kansal et al., 2025; Waghaye et al., 2023). Studies conducted in arid and semi-arid regions (Hassan et al., 2025b), Iraq (Hassan et al., 2025a), Australia (Khan et al., 2021), Bangladesh (Rana and Moniruzzaman, 2023), and Pakistan (Radwan and Alazba, 2023). Despite these advances, applications in Ethiopia remain limited and fragmented.

In Ethiopia, only a few studies have attempted to identify suitable sites for rainwater harvesting (Alene et al., 2022; Debebe et al., 2023b; Gebremedhn et al., 2023). Most of these investigations relied on conventional approaches that incorporated a limited number of biophysical parameters—typically slope, land use, soil type, and rainfall—to assess RWH potential (Alem et al., 2022; Debebe et al., 2023b; Roba et al., 2022). While Geographic Information Systems (GIS) and Multi-Criteria Decision-Making (MCDM) techniques have proven effective for spatial analysis and decision support, their predictive accuracy and scalability remain constrained by linear assumptions and subjective weighting schemes. In contrast, machine learning (ML) techniques offer a significant advancement by autonomously capturing complex, nonlinear, and high-dimensional relationships among diverse environmental and socio-economic variables (Chen et al., 2025; Hasan et al., 2025; Kabir et al., 2025).

Recent comparative and review studies further reinforce the shift from traditional MCDM approaches toward machine learning for environmental suitability assessments. Whereas MCDM techniques depend heavily on subjective weighting and linear assumptions, machine learning methods reduce such subjectivity by autonomously learning complex, nonlinear relationships from data. Başağaoğlu et al. (2022) demonstrated that ML-based hydrological and land-suitability models consistently outperform classical MCDM techniques in predictive accuracy and robustness, particularly when working with large multi-layer geospatial datasets. Similarly, Mariño et al. (2025) showed that ML algorithms exhibit markedly higher forecasting precision and generalization capacity than ARIMA statistical models.

Additional studies confirm the superior performance of ML in groundwater and RWH potential mapping. Vafadar et al. (2023) evaluated the effectiveness of logistic regression (LR), random forest (RF), support vector machine (SVM), and extreme gradient boosting (XGB) for identifying groundwater potential zones in the Tehran–Karaj plain, reporting particularly high predictive accuracy from tree-based ensemble models such as XGB and RF. Hasan et al. (2025) compared ML algorithms with the Analytical Hierarchy Process (AHP) for RWH potential assessment in the hilly regions of Bangladesh, finding that ML achieved significantly higher performance, with AUC values reaching 0.93. Likewise, a machine-learning-based framework assessing the financial viability of decentralized RWH systems demonstrated that ensemble approaches, particularly RF, provided optimal predictions for benefit–cost ratio estimation.

Collectively, these studies confirm that ML techniques offer a more objective, data-driven, and scalable framework for environmental decision-making than traditional approaches. Their superior capacity to model nonlinear, high-dimensional relationships makes machine learning particularly suitable for RWH suitability mapping in heterogeneous landscapes such as the Choke watershed.

Identifying potential RWH zones requires a multidisciplinary, data-driven approach that integrates hydrological, environmental, and socio-economic parameters. In complex regions such as the Choke watershed, traditional empirical or MCDM approaches often fail to capture nonlinear interactions among multiple factors. Machine learning algorithms, however, are capable of modeling high-dimensional and nonlinear relationships, producing more accurate and scalable suitability assessments.

In this study, we apply and compare three ensemble tree-based ML algorithms—Random Forest (RF), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost)—to identify potential RWH zones in the Choke watershed. Although all three algorithms are tree-based, they employ distinct ensemble learning strategies that offer complementary strengths. RF utilizes a bagging approach to reduce variance and enhance model stability, GB builds trees sequentially to minimize prediction errors, and XGBoost incorporates regularization and optimized gradient boosting to improve computational efficiency and generalization performance (Demir and Sahin, 2023; Kavzoglu and Teke, 2022; Rane et al., 2024). These algorithms were selected due to their proven performance in hydrological and RWH suitability mapping, particularly in data-limited semi-arid contexts.

The novelty of this research lies not merely in applying machine learning but in developing an integrated ML–GIS–RS framework specifically tailored for RWH suitability mapping in a complex highland watershed. Unlike previous studies that used limited biophysical variables and subjective MCDM weighting, this study incorporates twelve diverse environmental and socio-economic parameters within a data-driven ensemble-learning architecture. By systematically comparing three advanced tree-based algorithms (RF, GB, and XGBoost), the study provides the first comprehensive, high-resolution RWH suitability maps for the Choke watershed, offering both methodological innovation and applied value.

Accordingly, the objective of this study is to develop accurate, data-driven suitability maps for rainwater harvesting in the Choke watershed by comparing the performance of RF, GB, and XGBoost algorithms. The findings aim to support evidence-based planning, improve water resource management, and enhance agricultural resilience under increasing climate variability.

2. Material and methods

2.1. Study area

Choke is a large block of the highland mountain range, located in the central part of both the East and West Gojjam Administrative Zones in the Amhara National Regional State, northwestern Ethiopia. Choke Mountain serves as a water tower for the upper Blue Nile Basin, acting as the source for over 60 rivers and 270 springs (Teklu et al., 2023). The catchment area of Choke Mountain spans nine woredas, and the watershed is situated between 9° to 11°N latitude and 37° to 38°E longitude (Fig. 1). Diverse topographical features, ranging from plateaus and deep incised valleys to escarpments, plains, and gorges, characterize the region. Elevations within the area vary from 2800 to 4088 m above sea level. Due to the complexity of its topography, the region experiences significant local gradients in precipitation, temperature, and soil properties (Asrat et al., 2022). Choke Mountain is pivotal to the region's hydrology, being the primary source of the majority of the tributaries of the Blue Nile River. Among the four major rivers originating from the mountain are the Muga, Chemoga, Abeya, and Techma, in addition to several smaller tributaries (Abay) (Asrat et al., 2022). The mountain's abundant water resources support various nature-based tourism activities, including swimming, boating, hiking, and fishing. These water resources are also crucial for other purposes, such as electricity

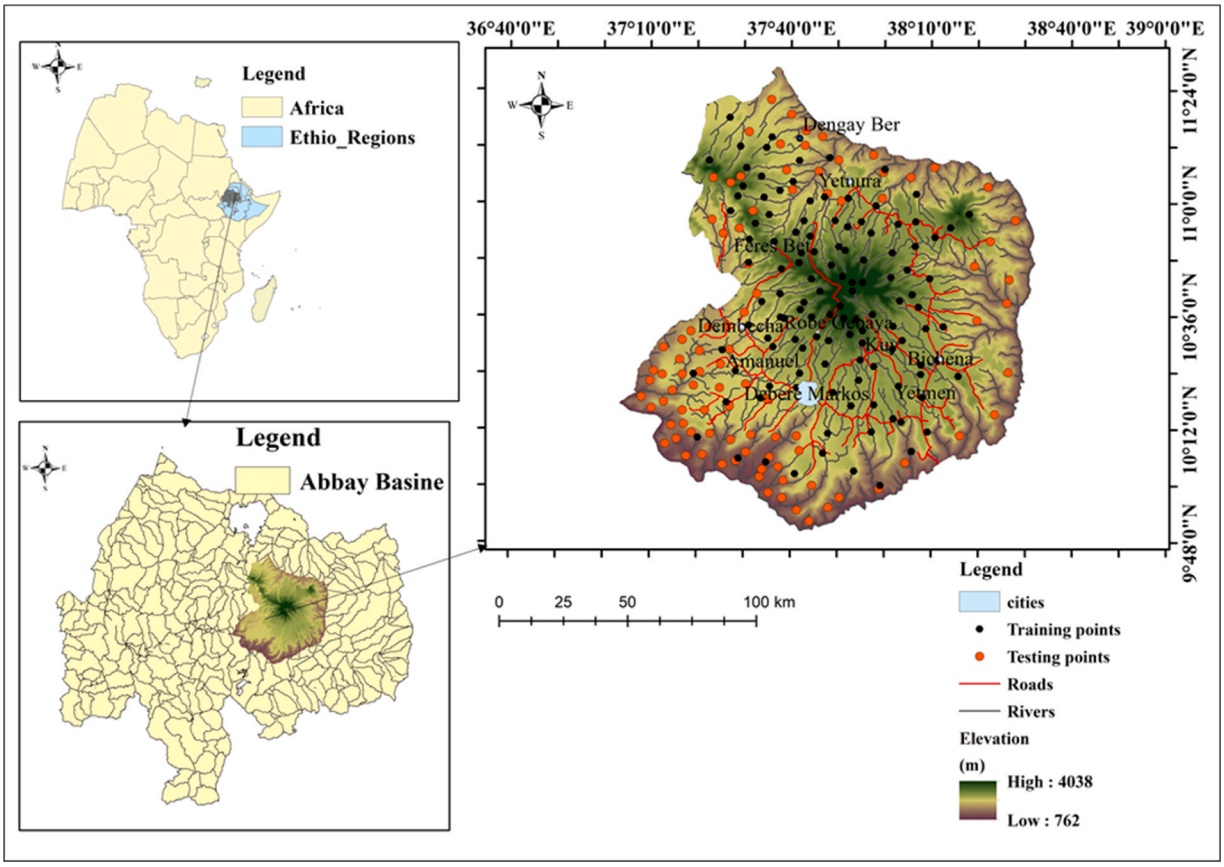


Fig. 1. Study area map and rainwater harvesting (RWH) inventory.

generation, drinking water, irrigation, and washing. Climatically, the Choke Mountain region spans six distinct climatic zones: Upper Kola, Lower Weyna Dega, Upper Weyna Dega, Lower Dega, Upper Dega, and Wurch. Rainfall in the region is highly variable, ranging from 600 to 2000 mm per year, with local differences due to the topographic gradients (Mesfin et al., 2023). The dominant soil types in the watersheds include Leptosols, Cambisols, Vertisols, Nitosols, Alisols, Luvisols, Andosols, and Phaozems, which support a variety of agricultural activities. Common crops cultivated in the region include sorghum, maize, teff, durum wheat, barley, chickpeas, and various pulses and potatoes, depending on the agro-ecological zone (Adane et al., 2015).

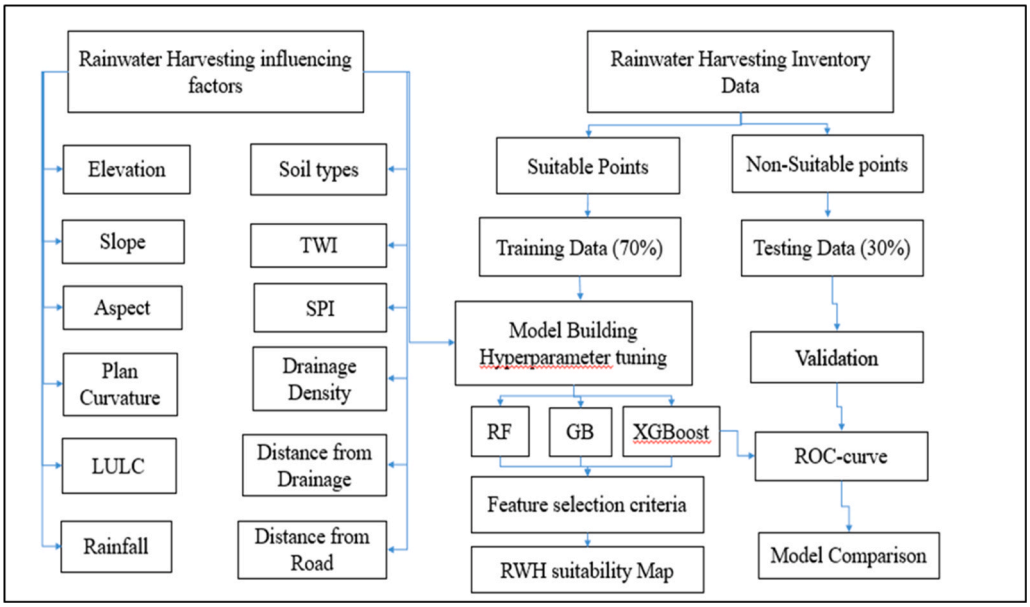


Fig. 2. Methodological workflow adopted in this study.

2.2. Methodology

The methodological framework adopted in this study is illustrated in Fig. 2 and consists of several sequential stages. First, a rainwater harvesting (RWH) inventory was prepared based on field observations and spatial data. Second, relevant RWH conditioning factors were identified and prepared. Third, a multicollinearity analysis was conducted to select suitable conditioning factors. Fourth, RWH suitability maps were developed using machine learning algorithms. Finally, the performance and accuracy of the models were evaluated using the Area Under the Receiver Operating Characteristic (AUROC) curve.

2.2.1. Rainwater harvesting inventory data

To develop reliable training and validation datasets for a rainwater harvesting (RWH) inventory, systematic field measurements and spatial verification are necessary (Chen et al., 2025; Hasan et al., 2025; Singha et al., 2025). In this study, a total of 250 RWH inventory points were collected through extensive field surveys and geospatial assessment. At each site, on-ground measurements included GPS-based location recording, soil texture and infiltration characteristics, surface slope conditions, land-use/land-cover features, proximity to runoff-contributing upslope areas, evidence of natural water accumulation, and observations of existing RWH structures or suitability indicators. Additional site attributes such as drainage conditions, accessibility, and local socio-environmental constraints were also documented.

Based on these criteria, 175 locations were classified as suitable (class value = 1) and 75 as non-suitable (class value = 0) for RWH. The final dataset was prepared in binary format and randomly divided into a 70:30 ratio for model training and testing, ensuring representative coverage across varying elevations, slopes, and land-use categories within the watershed.

2.2.2. Rainwater harvesting influencing factors

The conditioning factors used for rainwater harvesting (RWH) suitability mapping may vary depending on watershed characteristics and data availability. In this study, twelve key influencing factors were integrated to evaluate RWH potential within the Choke watershed. These factors include elevation, slope, aspect, plan curvature, Topographic Wetness Index (TWI), Standardized Precipitation Index (SPI), drainage density (Dd), distance from roads, distance from drainage, soil type, land use/land cover (LULC), and rainfall.

A Digital Elevation Model (DEM) obtained from ALOS-PALSAR was used to derive essential topographic parameters, including elevation, slope, aspect, plan curvature, drainage density, and TWI. The land use/land cover (LULC) map was obtained from the USGS GLOVIS database, while soil data were sourced from the Harmonized World Soil Database (HWSD v1.2). Rainfall data were collected from ten meteorological stations—Debre Markos, Rebu Gebya, Feres Bet, Dengay Ber, Dembecha, Amanueal, Yetmen, Bichena, Yetnura, and Kuy—located within and around the Choke watershed.

These influencing factors have been widely applied in previous RWH suitability studies due to their strong control over surface runoff generation, infiltration, and water accumulation processes. All spatial datasets were processed and analyzed in a Geographic Information System (GIS) environment and standardized to a raster resolution of 30 m to ensure spatial consistency (Table 1). Fig. 3 illustrates the spatial distribution of the influencing factors used as input variables for the machine learning models.

2.2.2.1. Elevation. Elevation affects runoff direction and rainfall distribution, which in turn affects the possibility of rainwater collecting. Accurate elevation data are essential for developing effective rainwater harvesting management strategies. Thus, elevation is a fundamental parameter in rainwater harvesting assessment (Al-Kindi and Alabri, 2024; Debebe et al., 2023b). Water generally follows the path of least resistance, typically along the steepest descent (Al-Kindi and Alabri, 2024). Higher elevations tend to receive increased rainfall. In contrast, lower-lying areas provide flattened surfaces conducive to the placement of storage tanks (Al Assi et al., 2023). This dataset was used to generate the elevation model of the Choke Mountain Watershed (Fig. 3a).

Table 1
Data type and sources used in this study.

No.	RWH-conditioning factor	Source	Notes on Classification & Relevance
1	Elevation	Alaska Satellite Facility (ASF)	Determines terrain variability, runoff potential, and water flow pathways.
2	Slope	Derived from DEM	Influences runoff velocity and infiltration; flatter slopes promote RWH suitability.
3	Curvature	Derived from DEM	Identifies convergent/ divergent flow zones that affect runoff accumulation.
4	Stream Power Index (SPI)	Derived from DEM	Represents erosive power; low-SPI areas favor RWH structures.
5	Topographic Wetness Index (TWI)	Derived from DEM	Highlights moisture accumulation zones important for RWH.
6	Aspect	Derived from DEM	Influences evapotranspiration and microclimate.
7	Rainfall	National Meteorological Agency (NMA)	Provides long-term rainfall climatology necessary for assessing catchment yield.
8	Drainage density	Derived from DEM	Indicates channel concentration; low drainage density areas improve storage feasibility.
9	Distance from river	Derived from stream network	Determines proximity to watercourses; buffer areas are ideal for certain RWH systems.
10	Soil	Harmonized World Soil Database (HWSD v1.2)	Classification: FAO soil texture classes (clay, loam, sandy loam). Relevance: Soil texture controls infiltration capacity, runoff generation, and water-holding potential—key criteria for RWH site selection.
11	Land use / Land cover (LULC)	USGS–GLOVIS (Landsat 8)	Classification: USGS Level-1 (agricultural land, forest, pastoral, urban, barren land, water bodies). Relevance: Determines land cover influence on runoff, surface roughness, and feasibility of installing RWH structures.
12	Distance from road	Local Road Network	Measures accessibility; essential for construction feasibility and maintenance of RWH systems.

2.2.2.2. Slope. Slope is another key parameter in rainwater harvesting, as it affects both the velocity and direction of surface water flow. Slope refers to the steepness or inclination of the terrain and is commonly derived from DEM data. Steeper slopes accelerate water runoff, potentially decreasing the availability of rainwater for collection. In contrast, gentler slopes (lower degrees) provide optimal conditions for RWH tanks, promoting water infiltration and storage (Aldiansyah and Wardani, 2023; McNabb et al., 2023). However, rainwater harvesting is influenced by additional factors such as land use, vegetation cover, and soil characteristics (Al-Kindi and Alabri, 2024). In this study, a slope map was generated using ArcGIS 10.4 software from the DEM. The slope data were then classified into five categories to facilitate rainwater harvesting analysis (Fig. 3b).

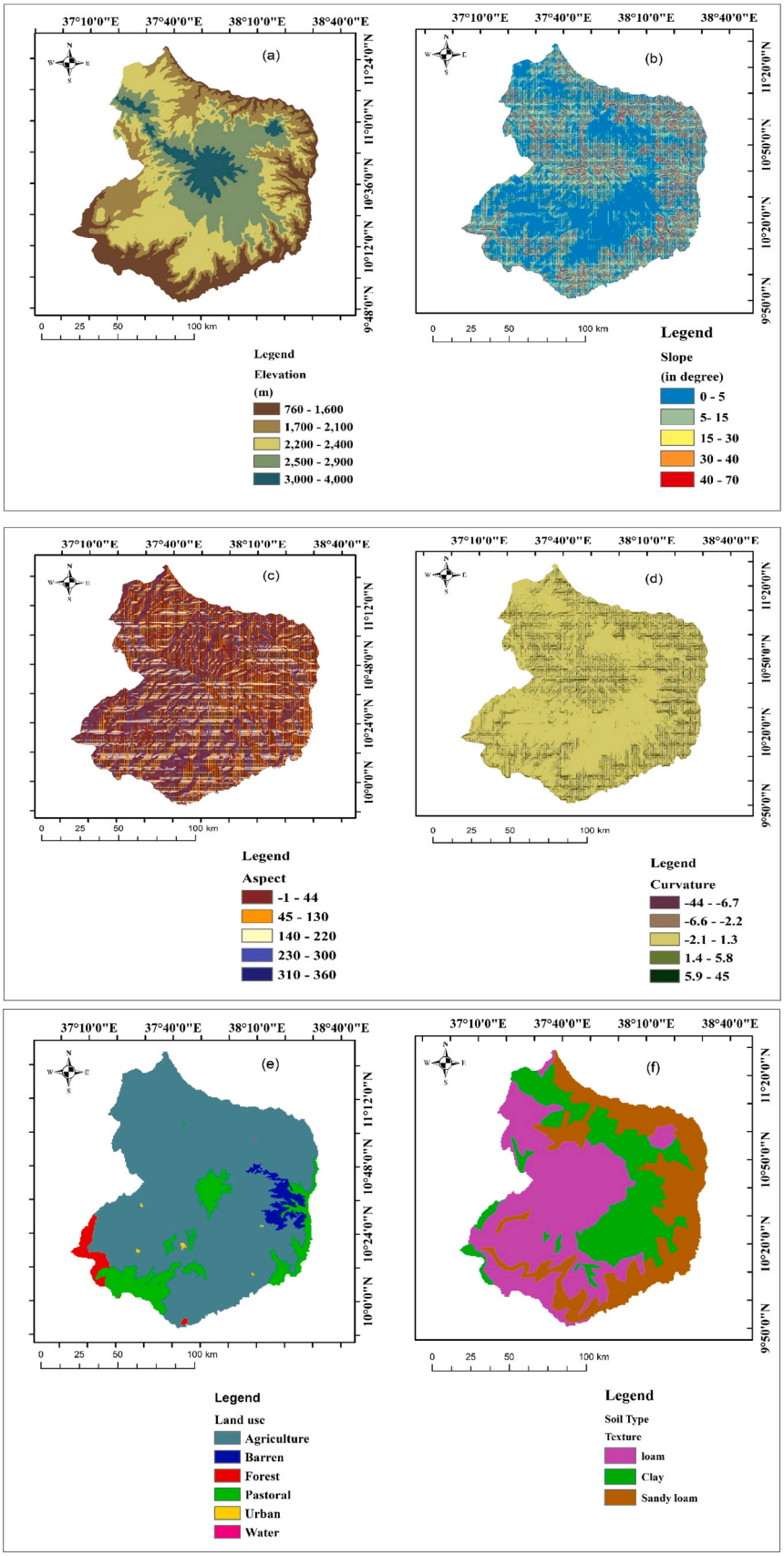


Fig. 3. Maps of rainwater harvesting factors: (a) elevation, (b) slope, (c) aspect, (d) curvature, (e) land use, (f) soil, (g) topographic wetness index (TWI), (h) drainage density (Dd), (i) Distance from drainage, (j) stream power index (SPI), (k) rainfall, and (l) Distance from road.

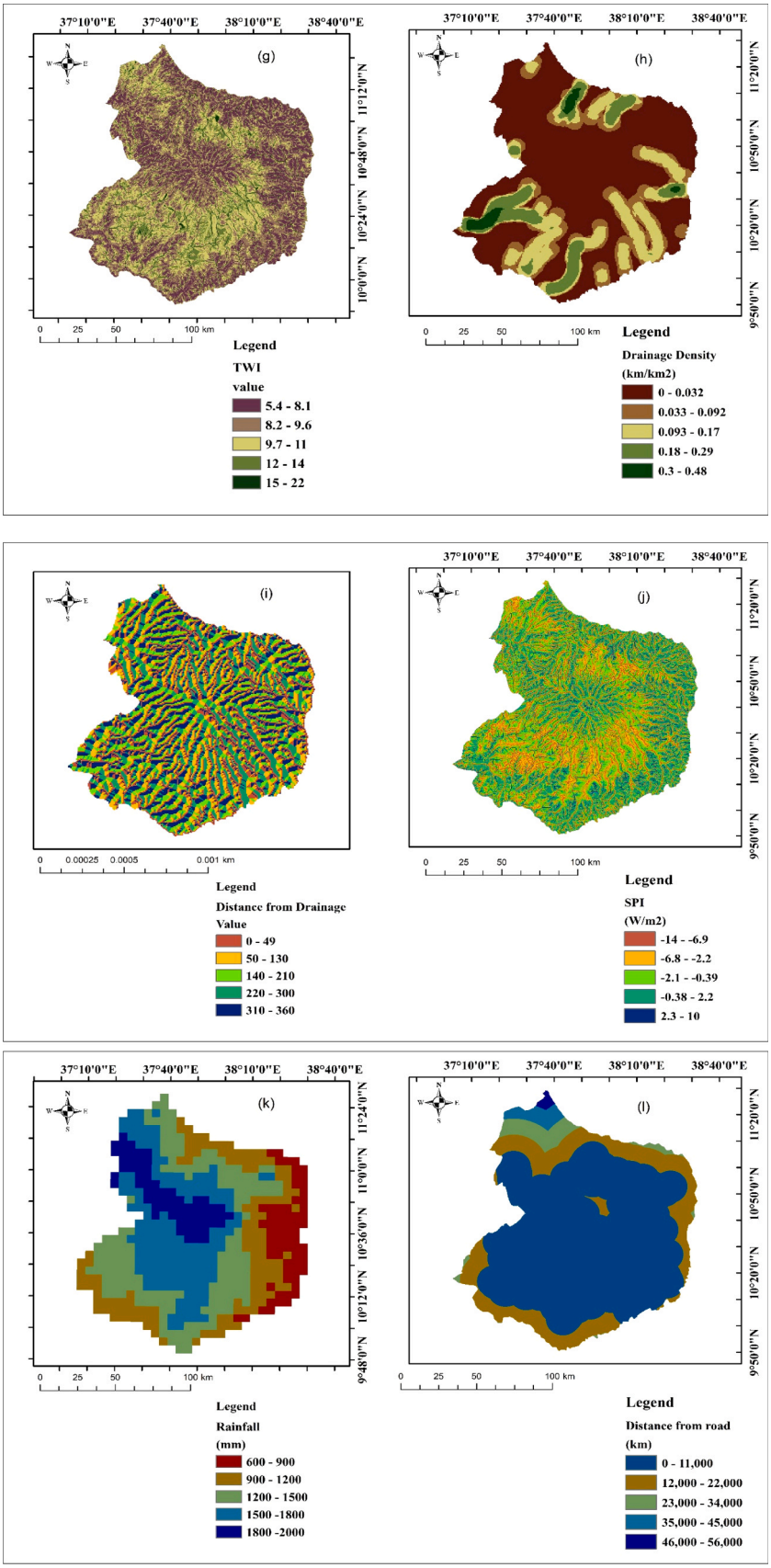


Fig. 3. (*continued*).

2.2.2.3. Aspect. Aspect is a key factor influencing the direction of water flow and related hydrological processes (Al-Kindi and Alabri, 2024). It refers to the compass direction that a slope faces and plays a significant role in determining microclimatic conditions, soil moisture distribution,

vegetation patterns, and overall watershed behavior (Al-Kindi and Alabri, 2024; Lu et al., 2025). Due to the impact of aspect on sunlight, exposure, and evapotranspiration rates, shady (north-facing) slopes tend to retain more moisture and experience less evaporation compared to sunny (south-facing) slopes. This can reduce the likelihood of waterlogging in shaded areas (Aldiansyah and Wardani, 2023). In this study, the aspect was derived from DEM data and categorized into five classes, each representing a specific cardinal or intercardinal direction (Fig. 3c).

2.2.2.4. Plan curvature. Plan curvature, also known as horizontal curvature, describes the curvature of the terrain surface perpendicular to the slope direction. It reflects the ability of the landscape to converge or diverge surface water flow, thereby influencing water accumulation and distribution patterns (Aydin and Iban, 2023; Gnann et al., 2025). The ground surface slopes upward in all directions when the positive curvature is high. This leads to faster water flow and an increased risk of erosion. Conversely, areas with high negative curvatures are concave, so the land surface slopes downward, leading to slower water flow and increased risk of ponding. Therefore, the likelihood of rainwater harvesting is generally inversely related to curvature—lower or more concave curvatures are associated with higher rainwater harvesting. In this study, plan curvature was derived from high-resolution elevation data using the ArcGIS 10.4 software. The resulting curvature map helps to identify terrain features that enhance or impede rainwater harvesting (Fig. 3d).

2.2.2.5. Land use/land cover. Land cover is directly linked to the proportion of infiltration and surface runoff in a given area (Hasan et al., 2025). Land use/land cover (LULC) is an important criterion in selecting and implementing rainwater harvesting (RWH) processes (Chen et al., 2025; Hasan et al., 2025). The type of land cover at a given place has a significant impact on the depth of runoff generated at that location. Densely forested areas are linked to increased rates of infiltration and interception, which lower runoff depths (Ahmad et al., 2024; Ibrahim et al., 2019). Conversely, bare land and water bodies tend to generate higher runoff depths, while vegetated areas generally produce lower runoff (Ammar et al., 2016). According to Yosef and Asmamaw (2015) and Maina and Raude (2016), bare land and water bodies are considered highly suitable for rainwater harvesting, whereas cultivated and grazing lands are moderately suitable. Natural or plantation forests and built-up areas, on the other hand, are categorized as marginally suitable.

In this study, land use and land cover were classified into six categories: agricultural land, barren land, forest, pastoral land, urban areas, and water bodies. Classification was performed using ArcGIS 10.4 software (Fig. 3e).

2.2.2.6. Soil types. Soil type is a fundamental factor in rainwater harvesting, as it affects the soil's ability to absorb water, regulate infiltration, and generate runoff. Different soil textures and compositions influence groundwater recharge rates and surface water movement, making soil data critical for rainwater harvesting assessments (Bhattarai et al., 2024; Gebreslassie et al., 2025). Soil maps can identify areas with high runoff potential or limited infiltration capacity. When combined with other topographic and land use data, such as slope and LULC, soil information improves the accuracy of rainwater harvesting (Al-Kindi and Alabri, 2024; Noori and Singh, 2023).

Clay, sandy loam, and loam are among the soil types examined in the present study (Fig. 3F). Clay soils often have lower infiltration rates but better potential for water retention, while sandy soils usually have higher infiltration rates but lower capacity for water retention. Understanding these soil types is crucial for assessing the suitability of locations for RWH and predicting potential challenges related to infiltration.

2.2.2.7. Topographic wetness index (TWI). The Topographic Wetness Index (TWI) is a quantitative measure used to estimate spatial variations in soil moisture and potential water accumulation across a landscape. It relates the local upslope contributing area to the slope, reflecting the influence of topography on water movement and accumulation (Latue and Rakuasa, 2023).

TWI is computed using the following formula:

$$\mathrm{TWI} = \ln\left(\frac{As}{\tan\beta}\right)$$

Where: As is the upslope contributing area per unit contour length, and β is the local slope angle.

Higher TWI values indicate areas that are more likely to be saturated or accumulate water, while lower values suggest better drainage. This makes TWI a vital parameter in identifying zones with high runoff potential and for rainwater harvesting. Higher TWI values suggest areas with lower slopes and a higher likelihood of water infiltration and saturation. These areas might be suitable for locating RWH structures, as they have the potential to capture and store more rainwater. In this study, TWI values were derived from the DEM using standard hydrological analysis tools (Saravanan and Abijith, 2022), and the spatial distribution of TWI is presented in (Fig. 3g).

2.2.2.8. Drainage density. Drainage density is a key parameter in rainwater harvesting, as it reflects the extent of drainage networks within a watershed and significantly influences surface runoff and rainwater harvesting. It is defined as the total length of all streams and rivers in a basin divided by the basin's total area (Hasan et al., 2025; Qasimi et al., 2024). This factor is especially useful for infrastructure planning, land-use regulation, and hazard mitigation (Marino et al., 2023). Higher drainage density typically results in faster rainwater runoff, potentially decreasing the amount available for harvesting. Conversely, lower drainage density may signify areas where water accumulates, potentially making them suitable for rainwater collection.

In the current study, drainage density was calculated using a Digital Elevation Model (DEM) in ArcGIS 10.4 software. The resulting drainage density map (Fig. 3h) provides valuable insight for rainwater harvesting and supports effective rainwater harvesting management planning (Al Assi et al., 2023; Marino et al., 2023).

2.2.2.9. Distance from drainage. The distance from drainage systems—such as rivers, streams, and canals—is another important variable in rainwater harvesting. Areas located closer to major drainage channels are typically more vulnerable to rainwater harvesting due to direct exposure to runoff and potential overflow during high rainfall events (Ammar et al., 2016; Ibrahim et al., 2019; Sayl et al., 2016). Poorly drained areas can lead to water accumulation and increased flood risk, especially when runoff exceeds the natural or engineered drainage capacity (Singh et al., 2017).

In this study, the Euclidean Distance tool in ArcGIS 10.4 software was used to generate a spatial layer representing the distance from drainage features within the study area (Fig. 3i).

2.2.2.10. Stream power index (SPI). The Stream Power Index (SPI) is widely used in rainwater harvesting to estimate the erosive power of surface runoff and channelized flow. It is a function of both drainage area and slope and quantifies the potential energy available to move water and sediment in a watershed (Al-Kindi and Alabri, 2024). It measures the intensity of flow accumulation and the capacity of the water to erode soil and reshape terrain.

In this study, SPI was calculated using DEM-derived data in ArcGIS 10.4. The following formula was used to compute SPI (Bhattarai et al., 2024):

$$\mathrm{SPI} = \mathrm{As}^{*} \tan(\beta)$$

Where:

- As = upslope contributing area (flow accumulation),
- Tan (β) = local slope gradient.

The resulting SPI map (Fig. 3j) highlights zones of potentially high erosion and flow energy, thereby contributing to a comprehensive rainwater harvesting.

2.2.2.11. Rainfall. Rainfall is crucial for selecting suitable locations for RWH and influences runoff depth and groundwater recharge capacity (Hasan et al., 2025). One important aspect affecting the likelihood of RWH is rainfall. Because of the increasing rainfall in this area, there were more resources available for collection and storage. In this study, rainfall data were acquired from ten meteorological stations (Debre Markos, Robu Gebaya, Feres Bet, Dengay Ber, Dembecha, Amanueal, Yetmen, Bichena, Yetnura, and Kuy) located within the Choke watershed. The kriging interpolation method was applied using ArcGIS 10.4 to generate spatial rainfall distribution maps, as it effectively captures spatial autocorrelation and provides statistically optimal rainfall estimates in regions with complex topography and limited station density (Qasimi et al., 2024). The resulting precipitation map is shown in Fig. 3k.

2.2.2.12. Distance from road. Distance from the road represents the proximity of potential rainwater harvesting (RWH) sites to existing road networks. Proximity to roads significantly influences factors such as transportation costs associated with the installation, maintenance, and management of RWH systems. Sites located closer to roads and residential areas are generally more favorable for RWH implementation than those situated farther away (Hasan et al., 2025). The suitability of land for RWH increases as its distance from roads decreases, as road accessibility facilitates the collection of essential resources and contributes to community convenience (Mohril et al., 2024). Moreover, proximity to roads enables easier transportation of construction materials required for the development of RWH structures.

Therefore, road proximity was used as one of the key criteria for identifying suitable RWH sites. The Euclidean Distance tool in the Spatial Analyst extension of ArcGIS 10.4 was applied to generate a road proximity map (Fig. 3l). The Raster Calculator tool was subsequently used to classify the study watershed into five suitability classes based on distance from roads. According to Khudhair et al. (2020), sites located within 0.5 km of a road were categorized as highly suitable, while those located more than 2 km away were classified as unsuitable for RWH development.

2.3. Machine learning (ML) algorithms

Machine learning (ML) algorithms were employed to capture complex, nonlinear relationships among hydrological, environmental, and anthropogenic variables that influence rainwater harvesting (RWH) suitability. In this study, three ensemble tree-based algorithms—Random Forest (RF), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost)—were selected due to their demonstrated robustness, high predictive accuracy, and suitability for geospatial hydrology applications (Chen et al., 2025; Vafadar et al., 2023). These models are particularly effective in handling multicollinearity, heterogeneous data distributions, and nonlinear relationships commonly observed in watershed-scale environmental datasets (Kansal et al., 2025; Nguyen et al., 2022; Rahman et al., 2019; Singha et al., 2025).

Although all three algorithms are tree-based, they employ distinct ensemble learning strategies that offer complementary strengths. RF utilizes a bagging approach to reduce variance and enhance model stability, GB builds trees sequentially to minimize prediction errors, and XGBoost incorporates regularization and optimized gradient boosting to improve computational efficiency and generalization performance (Demir and Sahin, 2023; Kavzoglu and Teke, 2022; Rane et al., 2024). This methodological diversity within the ensemble framework enables a comprehensive comparison of model behavior and predictive capability under varying environmental conditions.

Previous studies have consistently reported the superior performance of ensemble learning techniques over traditional single-model approaches such as logistic regression or classification and regression trees (CART), particularly in environmental susceptibility and RWH suitability mapping (Bhattarai et al., 2024; Hasnaoui et al., 2024; Ullah et al., 2025). Recent applications further confirm that RF-, GB-, and XGBoost-based models provide higher classification accuracy, reduced overfitting, and improved generalization across diverse climatic and topographic settings (Chen et al., 2025; Kansal et al., 2025; Singha et al., 2025). Consequently, these algorithms were selected to ensure robust, interpretable, and transferable predictions for RWH site suitability in the data-limited, semi-arid environment of the Choke watershed.

All conditioning factors were standardized to a 30 × 30 m resolution and preprocessed using ArcGIS 10.4 and Python 3.9. The dataset was thoroughly examined to ensure completeness and integrity. Missing values were systematically assessed across all layers, and void pixels detected in DEM-derived datasets were corrected using bilinear interpolation. Soil and land-use/land-cover attributes were cross-checked with their respective source datasets (HWSD v1.2 and Landsat-8). After these corrections, no remaining missing values were present in the final training dataset.

Model development and training were carried out in Python using the scikit-learn library (v1.1.3). Hyperparameters for each ML model were optimized using Randomized Search Cross-Validation, which explores a representative subset of the parameter space without the computational cost of exhaustive search (Rimal et al., 2024; Sharma et al., 2023). Model robustness and generalization performance were ensured through 10-fold cross-validation with five repetitions. The final predictive outputs were converted into spatial RWH suitability maps in ArcGIS 10.4, and model accuracy was evaluated using the Area Under the Receiver Operating Characteristic Curve (AUROC). Spatial patterns within the resulting suitability maps were further analyzed to assess the consistency and reliability of model predictions across different environmental settings within the watershed.

2.3.1. Random Forest (RF)

The Random Forest (RF) algorithm is a widely used machine learning technique that has shown strong performance in rainwater harvesting suitability and prediction (Hitouri et al., 2024). RF is an ensemble learning method that builds multiple decision trees and aggregates their outputs to improve predictive accuracy and reduce overfitting. For classification tasks, the final prediction is based on the majority vote of all decision trees, whereas for regression problems, the average of all individual tree predictions is used.

Each decision tree within the forest is constructed from a random subset of the training data and a random selection of input features, which introduces diversity among trees and enhances the model's robustness (Hitouri et al., 2024).

One of the primary advantages of RF is its high training speed and its ability to handle multicollinearity and nonlinear relationships among input features. Moreover, RF is relatively resilient to missing data and imbalanced datasets, maintaining strong predictive accuracy even in challenging data conditions (Bhattarai et al., 2024; Khosravi et al., 2019).

These characteristics make RF particularly suitable for flood susceptibility modeling, where environmental factors often interact in complex and nonlinear ways. Additionally, RF can provide insights into the relative importance of each input variable, which is useful for understanding the most influential rainwater harvesting suitability factors.

2.3.2. Gradient boosting model

The Gradient Boosting (GB) model is a powerful ensemble machine learning algorithm that has demonstrated high performance in both

classification and regression tasks, particularly on structured (tabular) datasets (Singha et al., 2022). In the context of rainwater harvesting suitability, GB model has been widely applied due to its ability to handle complex relationships and deliver accurate predictions.

GB model operates as an additive model, where decision trees are built sequentially. Each new tree attempts to correct the errors made by the previous ensemble of trees by fitting to the negative gradient of the loss function, which represents the direction of steepest descent (Singha et al., 2022). This iterative error-correcting process allows GB model to minimize prediction error effectively and improve model accuracy at each stage.

Unlike Random Forest (RF), which builds trees independently and aggregates their outputs, GBM constructs trees in a dependent sequence—where each subsequent tree is trained to reduce the residual errors from the preceding one (Hitouri et al., 2024). This stage-wise approach continues until a stopping criterion is met, such as reaching a specified number of trees or achieving minimal improvement in performance.

In this study, the GB model framework was utilized for rainwater harvesting suitability analysis due to its ability to reduce both bias and variance, offering robust generalization capabilities even in the presence of complex and nonlinear data relationships (Hitouri et al., 2024).

2.3.3. Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) is an advanced machine-learning algorithm that has gained significant popularity in solving regression, classification, and ranking problems. It builds on the traditional gradient boosting framework, enhancing it with advanced regularization, parallel computation, and efficient handling of missing data (Hitouri et al., 2024; Rahman et al., 2019; Singha et al., 2022).

XGBoost is an ensemble method that constructs decision trees sequentially. Each tree is trained to correct the residual errors of the previous tree, thereby minimizing a regularized objective function. This function includes both a loss term, which measures model accuracy, and a regularization term, which penalizes model complexity. This approach helps prevent overfitting and improves the model's ability to generalize to new data (Al-Kindi and Alabri, 2024).

One of the algorithm's key strengths is its capacity to manage missing data efficiently. During training, XGBoost automatically assigns missing values to the branch that results in the highest information gain, which allows it to maintain accuracy even with incomplete datasets.

2.4. Model validation and performance evaluation

Accurate validation of rainwater harvesting suitability is essential to ensure its predictive reliability and practical applicability. During the model evaluation stage, a combination of statistical metrics is used to assess performance comprehensively. Among these, the Area under the Receiver Operating Characteristic Curve (AUC-ROC), Positive Predictive Value (PPV), and Negative Predictive Value (NPV) are particularly important (Al-Kindi and Alabri, 2024; Bhattarai et al., 2024; Saravanan and Abijith, 2022).

The AUC quantifies the ability of a model to distinguish between suitable and unsuitable areas. It provides a single scalar value representing model performance across all classification thresholds. An AUC value closer to 1.0 indicates excellent discriminatory power, while a value near 0.5 suggests poor performance. The Receiver Operating Characteristic (ROC) curve, which plots sensitivity (true positive rate) against 1 - specificity (false positive rate), offers a visual assessment of the trade-offs between true and false positive classifications.

These metrics together offer a holistic view of model performance. High values of AUC indicate that the model is both accurate and reliable for rainwater harvesting suitability prediction. Consequently, the comparative analysis of these metrics across different machine learning models helps in identifying the best-performing approach for rainwater harvesting suitability in the study area (Chen et al., 2025; Hasan et al., 2025; Kansal et al., 2025).

2.5. Hyperparameter tuning

Hyperparameter tuning is an essential stage in machine learning model optimization, as it involves adjusting the non-trainable parameters that control the learning process and model behavior (Mohakud and Dash, 2022; Pillai et al., 2024). Proper tuning ensures that the models achieve optimal predictive performance and avoid issues such as overfitting or underfitting. Hyperparameters influence how classifiers interpret relationships within the dataset; therefore, they must be carefully fine-tuned prior to model training to enhance accuracy and generalization.

In this study, the GridSearchCV method was employed to identify the most effective combination of hyperparameters for each machine learning algorithm. GridSearchCV systematically evaluates a predefined set of hyperparameter values, training and validating the model for each possible combination. It then selects the configuration that yields the highest accuracy (Demir and Sahin, 2023; Hasanuzzaman et al., 2022). The optimized hyperparameters and their corresponding values that produced the best performance are presented in Table 2.

2.6. Feature selection criteria

Feature selection is an essential stage in creating effective and accurate machine learning (ML) models, as it enhances model interpretability, reduces overfitting, and improves computational efficiency. The inclusion of relevant and high-quality features strengthens predictive performance, whereas the presence of irrelevant or redundant variables can introduce noise, increase computational costs, and degrade model accuracy (Bilal et al., 2025; Dhanka et al., 2025). One of the major challenges in feature selection is multicollinearity, which occurs when one independent variable is a linear function of another (Ammar et al., 2016; Ibrahim et al., 2019). High multicollinearity undermines the reliability of regression coefficients, making the model statistically unstable and reducing its predictive validity (Hasanuzzaman et al., 2022). To evaluate the level of multicollinearity among predictor variables, statistical metrics such as variance inflation factor (VIF) and tolerance (TOL) are frequently used. The percentage of a predictor's variability that cannot be accounted for by other independent variables in the model is known as tolerance, and its values range from 0 to 1 (Dhanka et al., 2025; Theng and Bhoyar, 2024). A model is generally considered free from multicollinearity when TOL $>$ 0.1 and VIF $<$ 10 (Al-Kindi and Alabri, 2024).

Table 2
Tuned hyper-parameters.

Algorithm	Hyperparameters	Tuned value	GridSearchCV dictionary
RF	n_estimators	150	[5, 10, 25, 50, 150, 300]
	max_features	5	['sqrt', 'log2', None, 2, 5, 10]
	max_depth	None	[None, 5, 10, 15, 25]
	criterion	Gini	['gini', 'entropy', 'log_loss']
GB	learning_rate	0.02	[0.01, 0.02, 0.05, 0.005, 0.001]
	n_estimators	200	[5, 10, 25, 50, 150, 300]
	max_features	3	['auto', 'sqrt', 'log2', 2, 5]
	criterion	mse	['friedman_mse', 'squared_error', 'mse']
	max_depth	5	[3, 6, 9, 12, 15]
XGBoost	learning_rate	0.1	[0.1, 0.3, 0.5, 0.7, 1]
	gamma	0.6	[0.1, 0.3, 0.5, 1]
	max_depth	5	[3, 6, 9, 12, 15]
	alpha	0.02	[0.1, 0.01, 0.005, 0.001]
	max_features	2	[0.1, 0.3, 0.5, 1]

3. Results

3.1. Multicollinearity and feature selection

One of the most important steps prior to producing the rainwater harvesting suitability is to compute the importance scores of rainwater harvesting conditioning factors. To assess suitability associations among the conditioning factors, a multicollinearity test and pairwise Pearson's correlation analysis were performed. The results indicated that the highest variance inflation factor (VIF) value was 2.45, while the lowest tolerance (TOL) value was 0.41 (Table 3). Since all variables had VIF < 10 and TOL > 0.1, no multicollinearity was detected, confirming that all conditioning factors were suitable for model inclusion.

The variable importance rankings obtained from the three ensemble models—Random Forest (RF), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost)—are presented in Fig. 4. The results indicate that the rainfall, soil, and slope are the most critical factors for all used algorithms, followed by the elevation, drainage density, and land use. These findings align with previous studies, which consistently highlight the dominant roles of climatic input, infiltration characteristics, and terrain control in determining RWH suitability (Desalegn and Mulu, 2021; Diriba et al., 2024; Hasan et al., 2025; Singha et al., 2025).

In contrast, distance from drainage and distance from road demonstrated comparatively low importance. This reduced contribution is primarily attributed to their limited hydrological relevance in the context of small-scale rooftop and surface rainwater harvesting, where water collection depends more strongly on precipitation, terrain, and soil conditions than on proximity to rivers or transport infrastructure. Furthermore, both variables exhibit relatively homogeneous spatial patterns within the study area, offering less discriminatory power compared to highly variable topographic factors such as slope or curvature. Their consistently low ranking across RF, GB, and XGBoost models also indicates that the result is not an artifact of model bias but rather a reflection of genuine weak predictive relationships with RWH suitability.

The variables in the RF model with the highest importance were found to be rainfall, slope, soil, elevation, drainage density, and aspect, while variables with medium to lesser importance included land use, TWI, SPI, curvature, and distance from road.

The variables with the highest importance in the GB model are soil, rainfall, slope, drainage density, and elevation, while those with medium to low importance are TWI, SPI, land use, distance from road, and curvature.

The variables with the highest importance in the XGBoost model are the slope, soil, rainfall, drainage density, elevation, and land use, while those with medium to less importance are aspect, SPI, TWI, curvature, and distance from road.

3.2. Rainwater harvesting zone

The assessment of rainwater harvesting (RWH) suitability in the Choke watershed was conducted using three machine learning models—Random Forest (RF), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost)—to examine spatial variations in suitability influenced by geographic characteristics. Various classification approaches are available in the literature for categorizing suitability maps; however, the natural breaks (Jenks) classification method is widely applied because it effectively identifies class boundaries and enhances the interpretability of results (Abdo et al., 2024; Li et al., 2023). This method determines class intervals by minimizing within-class variance and maximizing between-class differences.

Accordingly, RWH suitability was classified into five categories—very low, low, moderate, high, and very high—using the natural breaks (Jenks) method in ArcGIS 10.4 (Fig. 5). The RF model indicates that very low, low, moderate, high, and very high suitability classes cover 2.15 %, 7.2 %, 57.3 %, 15.2 %, and 18.15 % of the study area, respectively. Similarly, the GB model estimates 3.2 %, 5.4 %, 42.42 %, 31.2 %, and 17.78 % of the area under very low, low, moderate, high, and very high suitability classes, respectively. For the XGBoost model, the corresponding proportions are 2.2 %, 6.2 %, 57.02 %, 20.02 %, and 14.56 % of the total watershed area.

A comparative summary of the spatial distribution of RWH suitability classes generated by each model is presented in Table 4, while the graphical comparison of suitability class proportions is illustrated in Fig. 6.

One of the most significant findings was that many areas in the northern and southwestern parts of the study area were located in high- and very high- rainwater harvesting zones. These zones are topographically characterized by gentle to moderate slope and soil composed of clay loam, low-lying terrain, and high Drainage density which intensify water accumulation during heavy rainfall events. The Eastern, Southwestern, and Central regions of the study area fall under very low and low suitable to moderately suitable. These zones are topographically characterized by steep slope, low Drainage density, sandy loam, and high terrain. The results agree with previous studies, such as those by Hasan et al. (2025), Waghaye et al. (2023), and Kabir et al. (2025), which indicate that the most suitable locations for rainwater collection are areas with a gentle to moderate slope and soil composed of loam or clay loam. These areas, which include bare land (Hasan et al., 2025), higher Drainage density (Gebremedhn et al., 2023), and those sufficiently far from lakes or rivers to be safe, offer the best conditions for the use of RWH technologies.

3.3. Evaluating the performance of different ML algorithms

To assess the predictive performance of the applied machine learning (ML) algorithms, Receiver Operating Characteristic (ROC) curve analysis was conducted. The ROC curves were used to calculate the Area under the Curve (AUC) values, which reflect each model's ability to distinguish between suitable rainwater harvesting and non- suitable rainwater harvesting areas. AUC values closer to 1.0 indicate higher model accuracy.

Among the tested algorithms, Gradient Boosting (GB) and Extreme Gradient Boosting (XGBoost) achieved the highest test accuracy (AUC = 0.97), followed closely by Random Forest (RF) with an AUC of 0.96. These findings highlight the superior performance of tree-based ensemble methods in rainwater harvesting suitability mapping. Fig. 7 shows the ROC curve of Gradient Boosting (GB), Extreme Gradient Boosting (XGBoost), and Random Forest (RF) respectively.

The effectiveness of Gradient Boosting (GB) and Extreme Gradient Boosting (XGBoost) in rainwater harvesting (RWH) suitability modeling is supported by previous research. For instance, Hasan et al. (2025) applied RF for RWH suitability assessment in Bangladesh and reported an AUC value of 0.93, which is slightly lower than the 0.97 achieved in the present study. Similarly, Madhuri et al. (2021) compared five machine learning (ML) algorithms—Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Adaptive Boosting

Table 3
Multicollinearity (MC) test of the Rainwater harvesting conditioning factors.

No.	Conditioning factor	VIF	TOL
1	Rainfall	2.45	0.41
2	Soil	2.42	0.41
3	Distance from drainage	1.08	0.93
4	Elevation	1.96	0.51
5	Slope	2.01	0.50
6	Drainage density	1.74	0.57
7	SPI	1.45	0.69
8	Curvature	1.32	0.76
9	TWI	1.58	0.63
10	Aspect	1.65	0.61
11	Land use/Land cover	1.72	0.58
12	Distance from road	1.05	0.95

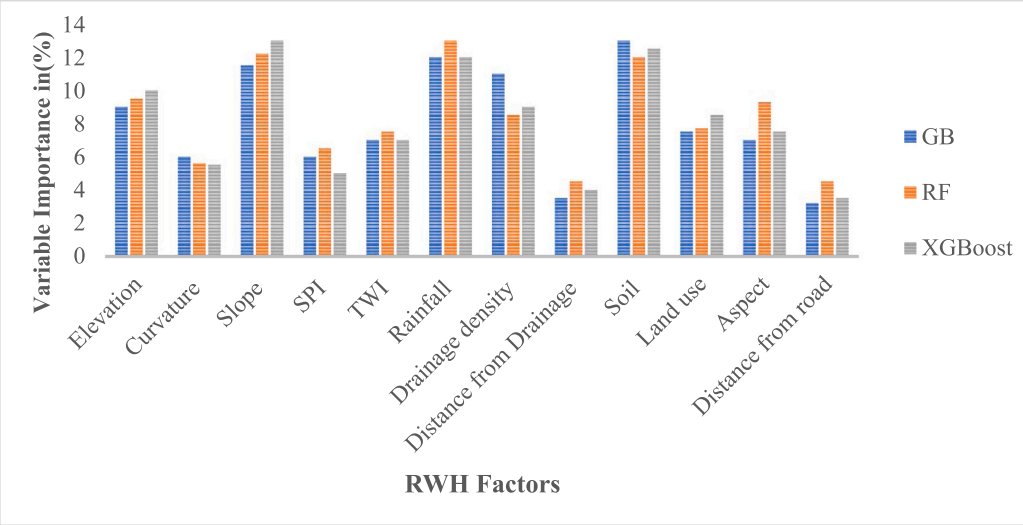


Fig. 4. Variable importance of the conditioning factors used in RF, GB, and XGBoost models.

(AdaBoost), and XGBoost—for the Greater Hyderabad Municipal Corporation (GHMC) region in India. Their results demonstrated that XGBoost outperformed the other algorithms, achieving the highest mean AUC of 0.83.

Additional studies further corroborate these findings. Sahin (2023) evaluated XGBoost, SVM, Random Forest (RF), and Logistic Regression for flood susceptibility mapping (FSM) and concluded that XGBoost yielded the most accurate predictions. Similar, Saravanan and Abijith (2022) assessed several ML algorithms—including GBM, XGBoost, Rotation Forest (RTF), SVM, and Naïve Bayes (NB)—for flood susceptibility mapping in northern coastal Tamil Nadu. Their results indicated that GB and XGBoost achieved the best predictive performance, with AUC values of 0.92 and 0.91, respectively.

Overall, the results of the present study align with existing literature, reinforcing that ensemble-based ML models—particularly GB and XGBoost—are highly effective and reliable for RWH suitability mapping.

4. Discussion

This study successfully integrated machine learning (ML) algorithms with geospatial techniques to assess rainwater harvesting (RWH) potential in the Choke watershed, Ethiopia. The findings provide critical insights for sustainable water management in a region increasingly affected by water scarcity, population growth, and high dependence on rainfed agriculture. Traditional RWH assessments that rely on ground-based observations often suffer from limited spatial coverage and poor temporal resolution (Hitouri et al., 2024). In contrast, modern ML approaches such as RF, GB, and XGBoost have demonstrated high accuracy, robustness, and efficiency in modeling complex hydrological processes (Chen et al., 2025; Hasan et al., 2025; Singha et al., 2025).

Feature importance analysis identified rainfall, soil, slope, drainage density, elevation, Topographic Wetness Index (TWI), and SPI as the most influential variables in this study. The strong influence of rainfall and soil is consistent with the climatic and agricultural realities of the Choke watershed. The area is characterized by a highly seasonal rainfall regime, where most precipitation occurs during a short rainy period. Since smallholder farmers rely heavily on rainfed crop production, interannual rainfall variability often results in crop failure and heightened vulnerability to drought. Consequently, locations receiving consistent or relatively higher rainfall naturally offer greater potential for effective rainwater harvesting.

Similarly, the widespread presence of loam and clay-loam soils across the watershed strongly enhances RWH suitability. These soil types have moderate to high water-holding capacity, enabling them to retain harvested water in the root zone for longer periods—an essential advantage for sustaining crops during dry spells. In contrast, sandy soils, which exhibit rapid infiltration and low moisture retention, were consistently identified as less suitable. These findings reinforce previous studies highlighting the pivotal role of local climatic and edaphic conditions in determining the viability of RWH technologies (Gebremedhn et al., 2023; Hasan et al., 2025; Kabir et al., 2025).

Topographic factors such as slope also played an important role. Steep slopes reduce feasibility due to high runoff velocity and greater construction costs for RWH structures, whereas gently sloping and flat terrain—common in the southern, northern, and western parts of the watershed—enable efficient surface runoff collection and cost-effective construction of ponds, trenches, and other structures. Similar observations were documented in previous RWH suitability assessments conducted in Ethiopia and other semi-arid regions (Gebremedhn et al., 2023; Moges et al., 2011; Roba et al., 2022; Tolossa et al., 2020).

Land use/land cover (LULC) patterns further influenced suitability outcomes. Bare lands and areas surrounding water bodies were classified as highly suitable because they allow efficient runoff accumulation and can be enhanced through small reservoirs or diversion weirs. Conversely, forested and built-up areas were moderately suitable due to conservation requirements and infrastructure limitations. These observations align with earlier research demonstrating the importance of selecting LULC types that maximize surface runoff storage while minimizing ecological disruptions. Gebremedhn et al. (2023) and Yegizaw et al. (2022) also reported that utilizing bare lands and water bodies for rainwater collection is advantageous for sustainable and efficient land use. These land use types are capable of capturing and channeling larger volumes of surface runoff from small catchments compared to other LULC classes (Doost et al., 2024).

Water bodies are particularly suitable for RWH development because they naturally facilitate water storage and can be further enhanced through the construction of hydraulic structures such as small dams, diversion weirs, and floodwater retention systems. Similar suitability classifications have been applied in previous studies by Ejegu and Yegizaw (2020), Gebremedhn et al. (2023), and Yegizaw et al. (2022), further validating the classification approach used in this study.

The effectiveness of RWH is particularly high in areas with greater drainage density (Dd), where multiple stream networks enhance surface runoff potential (Gebremedhn et al., 2023; Hasan et al., 2025). In addition to hydrological factors, accessibility also plays a crucial role in determining RWH suitability. Sites located near roads and residential areas are generally preferred over remote locations because they facilitate easier access for construction, maintenance, and utilization

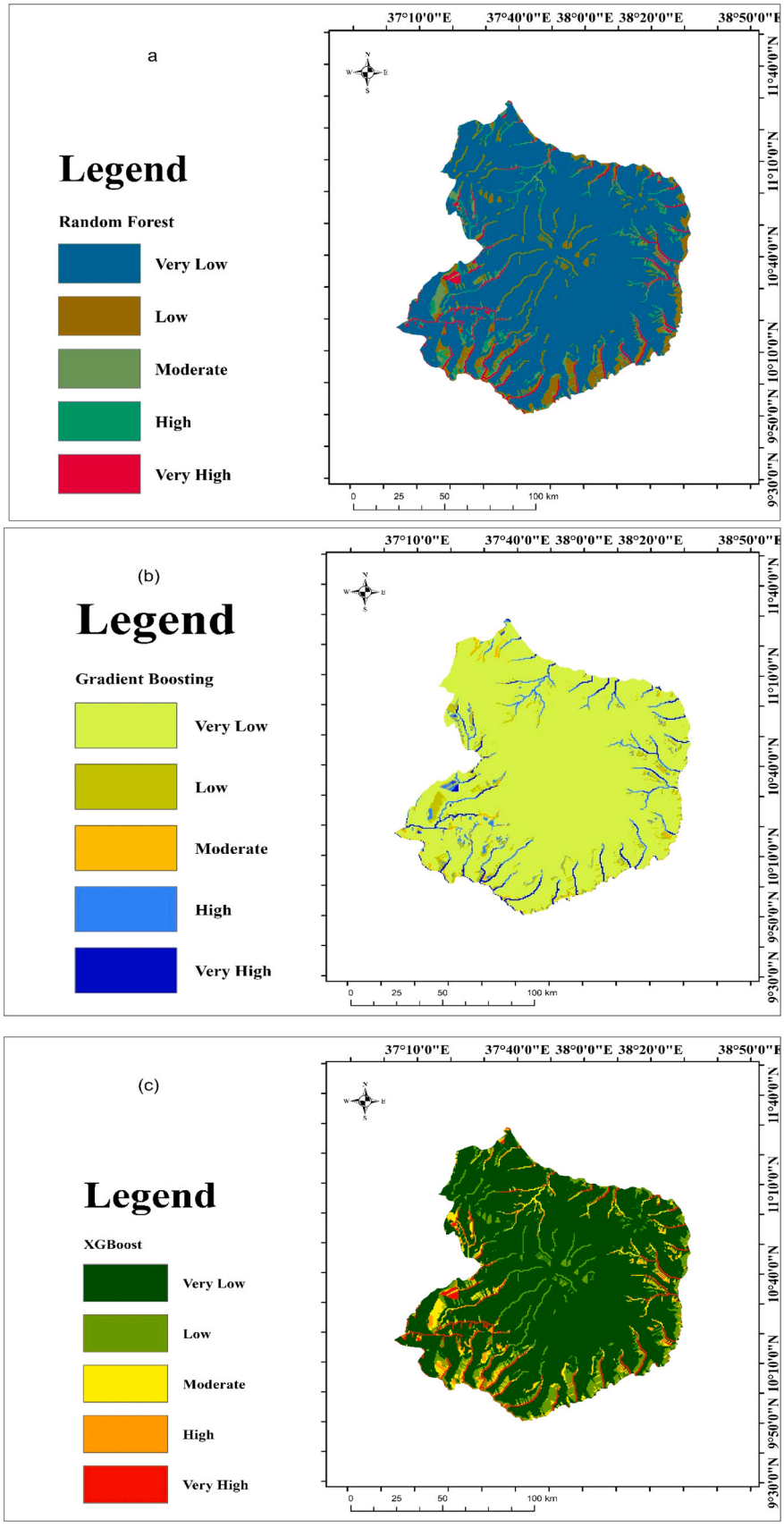


Fig. 5. Rainwater harvesting suitability: (a) RF, (b) GB, and (c) XGBoost.

Table 4
Areal coverage of water harvesting suitability in RF, GB, and XGBoost models.

Models	very low	Low	Moderate	High	Very High
RF	2.15 %	7.20 %	57.30 %	15.20 %	18.15 %
GB	3.20 %	5.40 %	42.42 %	31.20 %	17.78 %
XGBoost	2.20 %	6.20 %	57.02 %	20.02 %	14.56 %

(Al-Shabeeb et al., 2023; Tolossa et al., 2020). Proximity to road networks enhances the feasibility of transporting construction materials and equipment for developing RWH structures while also improving community access and participation (Gebremedhn et al., 2023). Therefore, road proximity was considered one of the key criteria in identifying convenient and cost-effective RWH sites in this study.

The study's results carry important practical implications. Identifying highly suitable zones for RWH provides actionable guidance for both farmers and planners. For smallholder farmers, these zones highlight areas where simple, low-cost in-situ RWH practices—such as contour bunds, infiltration trenches, or farm ponds—can be implemented with high success rates. Improved soil moisture availability can substantially increase crop yields, reduce drought-related crop failures, and enhance household water security. For planners, development agents, and water managers, the spatial suitability maps offer a decision-support tool for targeting investments, establishing community ponds, designing small-scale irrigation schemes, and integrating RWH into climate resilience programs. Prioritizing development in high-suitability zones increases cost-effectiveness, reduces failure risks, and ensures long-term sustainability of RWH interventions.

In terms of model performance, all three ML algorithms demonstrated strong predictive capability with AUC values exceeding 0.90. However, GB and XGBoost—both tree-based ensemble models—outperformed algorithms, consistent with prior studies (Al-Kindi and Alabri, 2024; Bhattarai et al., 2024; Jahanbani et al., 2024). Setianto et al. (2025), in a study on flood susceptibility in the Musi River basin, Hyderabad, India, found that XGBoost outperformed logistic regression, support vector machine (SVM), K-nearest neighbors (KNN), and AdaBoost—aligning closely with the findings of this study. Similarly, Wang et al. (2022) demonstrated that XGBoost outperformed traditional regression models in predicting mortality from moderate-to-severe traumatic brain injury. Demir and Şahin (2022) applied SVM, RF, and XGBoost for liquefaction prediction using a genetic algorithm-based feature selection method and reported that XGBoost provided the highest prediction accuracy, including in flash flood susceptibility modeling.

Although this study provides an important contribution to rainwater harvesting (RWH) suitability mapping using ML and geospatial techniques, several limitations should be acknowledged. The analysis relied on 250 training samples, which may constrain the generalizability of the models and increase sensitivity to class imbalance. While the models performed well, the limited sample size may not fully represent the spatial heterogeneity of the Choke watershed. Future research should expand the inventory of suitability points and apply data-augmentation techniques such as SMOTE to balance classes and improve model robustness. In addition, adopting k-fold cross-validation rather than a single train–test split would provide a more stable estimate of model performance.

Uncertainties in the spatial and temporal resolution of environmental datasets may affect prediction accuracy. Higher-resolution remote sensing data, updated climatic records, and additional field validation would help reduce these uncertainties.

Although the Gradient Boosting (GB) and Random Forest (RF) models demonstrated strong predictive accuracy, their "black-box" nature limits understanding of how individual variables influence RWH potential. Improving model transparency through interpretable ML approaches—such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations)—could provide deeper insights into the relationships among hydrological, topographic, and land use factors (Alsumaiei, 2025; Başağaoğlu et al., 2022). Additionally, future research could explore hybrid or deep learning (DL) models that can capture more complex nonlinear relationships and improve predictive performance (Parviz et al., 2023).

This study primarily focused on biophysical and hydrological variables and did not explicitly incorporate socio-economic factors such as land ownership, water use practices, and community participation, all of which significantly influence the practical implementation of RWH systems. Future research should integrate socio-economic datasets and participatory approaches to ensure that technically suitable sites are also socially and economically viable.

The impacts of climate change present a major challenge for sustainable RWH development. Changing precipitation patterns, increased temperature, and higher evapotranspiration rates can alter runoff generation and water storage potential (Field et al., 2018). These changes complicate the identification of long-term suitable RWH sites when relying solely on historical data. Future research should employ climate-scenario-based modeling to assess the resilience of RWH systems under projected rainfall variability and extreme weather events.

Future research on rainwater harvesting (RWH) suitability should focus on enhancing model precision, interpretability, and practical relevance. Integrating high-resolution spatial and temporal datasets will improve the accuracy of RWH suitability assessments by capturing finer variations in topography, soil texture, and rainfall distribution. Combining physically based hydrological models with advanced

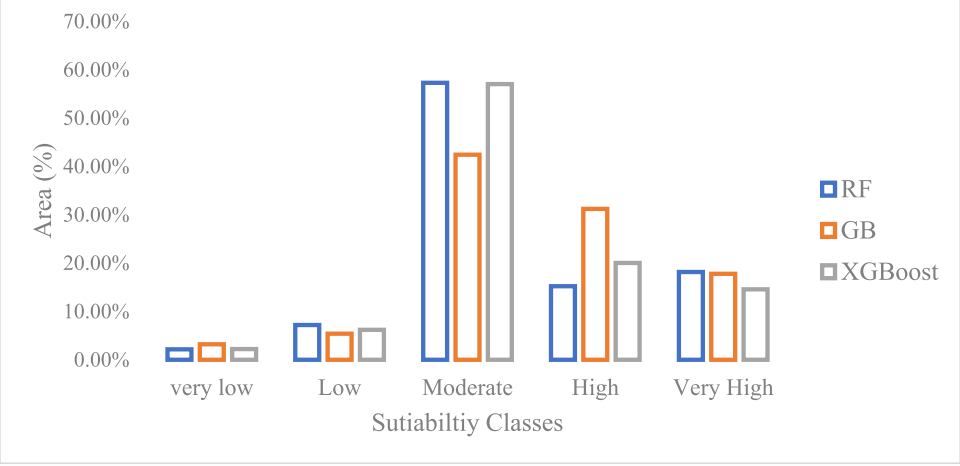


Fig. 6. Area coverage of rainwater harvesting suitability classes in RF, GB, and XGBoost models.

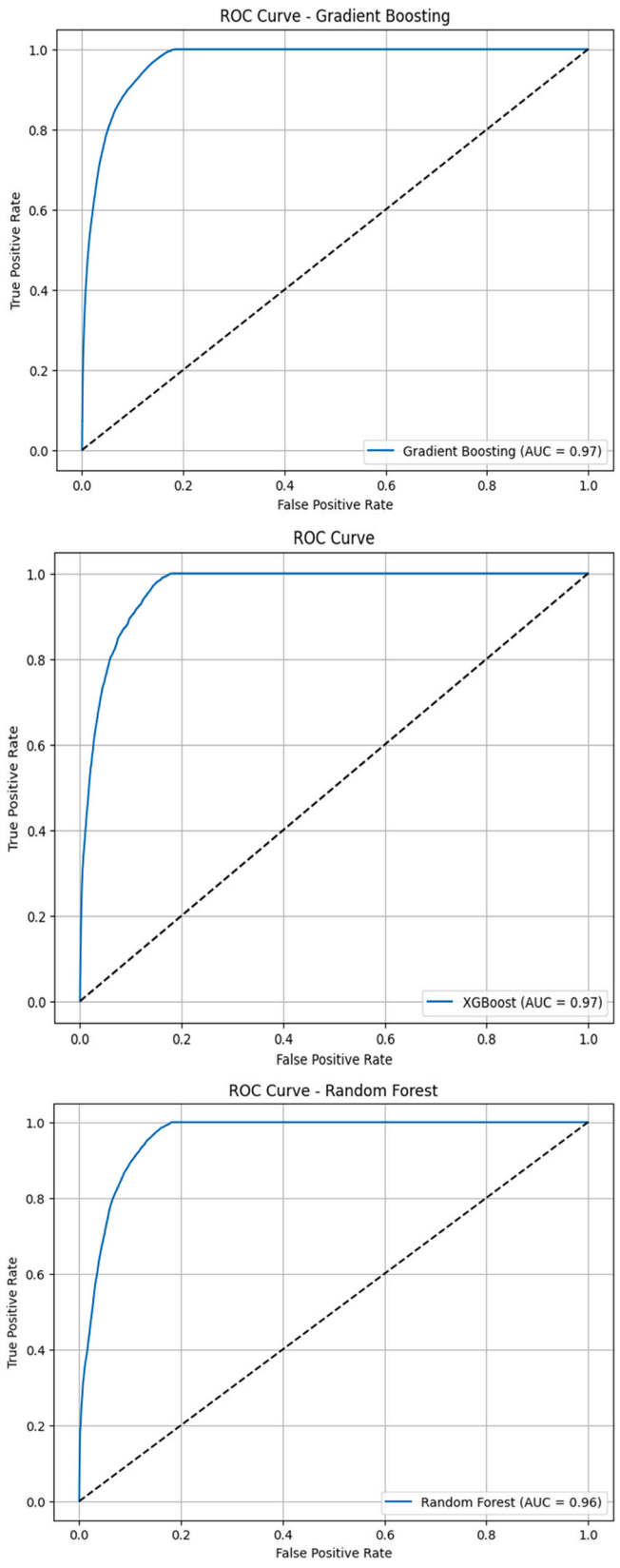


Fig. 7. ROC curve of the GB, XGBoost, and RF models validation.

machine learning (ML) and deep learning (DL) techniques can enhance both predictive performance and interpretability, allowing for a more comprehensive understanding of the hydrological and environmental processes influencing RWH potential. Moreover, incorporating socio-economic and institutional factors—such as land ownership, community participation, and policy frameworks—into spatial modeling will ensure that technically suitable sites are also feasible and sustainable for implementation. Finally, evaluating RWH suitability under different climate-change scenarios using downscaled regional climate models will provide critical insights into the long-term resilience and adaptability of RWH systems amid increasing rainfall variability and extreme weather conditions.

5. Conclusions

Identifying suitable sites for rainwater harvesting (RWH) by considering multiple influencing factors is critical for addressing water scarcity in the Choke watershed. In this study, twelve key factors—including slope, soil texture, land use/land cover (LULC), rainfall, Topographic Wetness Index (TWI), drainage density (Dd), Standardized Precipitation Index (SPI), aspect, curvature, elevation, distance from drainage, and distance from roads—were systematically evaluated using an integrated Geographic Information System (GIS) and machine learning (ML) approach. The Gradient Boosting (GB) and Extreme Gradient Boosting (XGBoost) models achieved the highest test accuracy (0.97), followed closely by Random Forest (RF) at 0.96, demonstrating the robustness of ensemble-based ML algorithms for RWH suitability mapping.

The spatial outputs reveal that RF suitability classes of very low, low, moderate, high, and very high cover 2.15 %, 7.2 %, 57.3 %, 15.2 %, and 18.15 % of the watershed, respectively. The GB model shows 3.2 %, 5.4 %, 42.42 %, 31.2 %, and 17.78 %, while XGBoost predicts 2.2 %, 6.2 %, 57.02 %, 20.02 %, and 14.56 % of the area across the same classes. These results provide clear, quantitative guidance for prioritizing areas for RWH interventions, including check dams, ponds, and in-situ conservation measures.

The generated suitability maps are valuable decision-support tools for policymakers, water resource managers, and local communities, enabling targeted investments that maximize hydrological and socio-economic benefits. By identifying high-potential zones, these outputs can strengthen regional water security, mitigate dry-season shortages, and support climate-resilient agricultural planning. The methodological framework is replicable and adaptable to other topographically complex, data-limited regions, offering broader international applicability. Future work integrating higher-resolution datasets, additional socio-economic factors, and climate scenarios can further enhance model precision, robustness, and long-term relevance. Overall, this study provides actionable insights and a robust framework for sustainable RWH planning, contributing to improved water resource management in the Choke watershed and similar environments worldwide.

Authors' contributions

TA has made considerable contributions in designing the study, writing original manuscript, data acquisition, data collection, analysis, interpretation, and manuscript writing; KB, and SA have made a significant contribution in designing and analysis of data, in commenting, suggesting ideas, and editing the manuscript.

All authors read and approved the final manuscript.

CRediT authorship contribution statement

Tade Mule Asrade: Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Sintayehu Adefires Abebe:** Writing – review & editing, Visualization, Supervision, Conceptualization. **Kassahun Birhanu Tadesse:** Writing – review & editing, Visualization, Supervision, Conceptualization.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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